



VEL TECH HIGH TECH

Dr. RANGARAJAN Dr. SAKUNTHALA ENGINEERING COLLEGE

An Autonomous Institution

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INNOVATIVE ASSIGNMENT-I

FACULTY NAME:	Mrs.SUGANYA DEVI S	FACULTY ID:	HTS1801
COURSE CODE:	21HC44T	COURSE NAME:	OPERATING SYSTEMS
YEAR/SEM:	II/IV	SEC:	C

SAMPLE ASSIGNMENT FORMAT

1. Analyze why paging reduces external fragmentation but may increase internal fragmentation.

Paging removes the need for contiguous memory, eliminating external fragmentation, but fixed-size pages can leave unused space inside frames, causing internal fragmentation.

1. External Fragmentation – Why It Occurs

External fragmentation happens when free memory is split into small non-contiguous blocks, even though total free memory is sufficient.

Without Paging (Contiguous Allocation)

Main Memory | P1 | Free | P2 | Free | P3 |

- Free spaces exist
- But no **single large block** is available
- New process cannot be loaded.

➔ **External fragmentation occurs**

2. Why Paging Eliminates External Fragmentation

Paging Concept

- Memory is divided into **fixed-size frames**
- Processes are divided into **fixed-size pages**
- Pages can be placed **anywhere in memory**

With Paging (Non-contiguous Allocation)

Main Memory (Frames)

| P1 | P2 | P3 | P1 | P2 | P3 |

- Pages of a process are stored in **different frames**
- No need for **contiguous free space**

➔ **External fragmentation is eliminated**

3. Internal Fragmentation – Why It Occurs in Paging

Internal fragmentation occurs when allocated memory is slightly larger than required, leaving unused space inside the allocated block.

Cause in Paging

- Page and frame sizes are **fixed**
- Last page of a process may not fully occupy a frame.

Example

- Page size = **4 KB**

- Process size = **10 KB**

PagesNeeded=3

Total MemoryAllocated=12 KBUnusedMemory=2 KB

➡ **Internal fragmentation occurs inside the last frame**

4.Diagram Showing Internal Fragmentation

Frame (4 KB)

| Used | Used | Used | Free | ↑

Internal Fragmentation

- Unused space **cannot be shared** with another process.

5.Cause–Effect Analysis

Paging uses fixed-size blocks

↓No need for contiguous memory

↓External fragmentation removed

↓

Unused space in last page

↓Internal fragmentation introduced

6.Comparison Table (Innovative Analysis)

Aspect	External Fragmentation	Internal Fragmentation
Occurs	Between memory blocks	Inside allocated block
Cause	Contiguous allocation	Fixed page size
Paging effect	Eliminated	Increased
Memory reuse	Difficult	Limited

7.Real-Life Analogy (Bus Seating)

- **Paging** → Seats of equal size in a bus
- A passenger group may not fill the last seat fully
- Empty space **inside the seat** cannot be used.

➡ That empty space = **internal fragmentation**

Conclusion

- Paging **eliminates external fragmentation** by allowing processes to be stored in **non-contiguous memory locations**.However, because memory is allocated in **fixed-size pages**, the unused space in the last page of a process leads to **internal fragmentation**.
- Thus, paging trades **external fragmentation for internal fragmentation**.

INNOVATIVE ASSIGNMENT-1 PROBLEM STATEMENTS

BATCH NO	STUDENT NAME	PROBLEM STATEMENTS	CO	K LEVEL
1	RONIK SAMUEL S ROSHAN S SAI PRASAD M	A)Apply the computer system model (User → Application → OS → Hardware) to explain how an online exam system works.	CO1	K3
		B)Analyze the impact of poor process scheduling on system performance in a real-time environment. Explain how missed deadlines, increased waiting time, and low CPU utilization can affect critical applications, and suggest strategies to improve scheduling efficiency.	CO2	K4
		C)Apply the concept of main memory to explain how an Android application is loaded and executed.	CO3	K3
2	ROSHINII B SAJITHA SREE R SANDHIYA P	A) How does the balance between CPU speed, memory capacity, and I/O performance affect the overall efficiency of a computer system? Explain with a real-world example.	CO1	K3
		B) Compare and analyze processes and threads in terms of resource usage, execution speed, and system performance. Explain how threads improve efficiency in multitasking environments while highlighting the differences in isolation, memory sharing, and context switching overhead.	CO2	K4
		C)Demonstrate swapping using a real-life analogy (classroom/library) and explain its role in memory utilization.	CO3	K3
3	SAI RAM R SAKTHIVEL M SANJAI KUMAR S B	A) Evaluate how a smartphone operating system achieves the goals of resource management and multitasking, and explain their impact on real-time user experience.	CO1	K3
		B)Analyze how scheduling requirements differ between desktop, server, and mobile operating systems, focusing on fairness and throughput.	CO2	K4
		C)Apply contiguous memory allocation to allocate memory for a given set of processes and draw the memory layout.	CO3	K3

4	SANJEEV YADAV	A) Demonstrate how an operating system performs process management when multiple applications are running simultaneously.	CO1	K3
	S	B) Analyze the advantages and disadvantages of preemptive and non-preemptive CPU scheduling. Explain how each approach affects system responsiveness, fairness, context switching overhead, and suitability for different types of applications.	CO2	K4
	SANTHOSH H SANTHOSH S	C) Apply the first-fit memory allocation technique to a given set of processes and memory blocks, and analyze the resulting internal fragmentation.	CO3	K3
5	SARAN M G	A) Apply the concept of multiprogramming to a college computer laboratory and analyze how it improves CPU utilization and system efficiency.	CO1	K3
	SARAVAN SIDDHARTH R	B) Examine the critical-section problem in operating systems and analyze why improper handling can lead to race conditions. Explain the conditions necessary for a race, and discuss synchronization mechanisms that prevent inconsistent data access.	CO2	K4
	SASI KARAN M	C) Apply best-fit and worst-fit allocation for the same process set and compare the results.	CO3	K3
6	SARAYU	A) Illustrate the evolution of operating systems by mapping each stage to a real-world computing device (mainframe, PC, mobile).	CO1	K3
	NANDHINI S	B) Analyze the role of synchronization hardware instructions, such as Test-and-Set and Compare-and-Swap, in solving the critical-section problem. Explain how these instructions ensure mutual exclusion, prevent race conditions, and improve system reliability in concurrent environments.	CO2	K4
	SATHVIKA D SHAMITHA M	C) Apply the concept of paging by dividing a given logical address into a page number and page offset. Explain how this division helps in mapping logical addresses to physical memory frames and prevents external fragmentation.	CO3	K3
7	SETHURAMAN B	A) Apply any two operating system services—such as file management and I/O management—to explain the steps that occur when a user creates, edits, and saves a file. Analyze how these services ensure data integrity, efficient storage, and user convenience.	CO1	K3
	SHAIK ARMAN BASHA SHAM S	B) Analyze classical synchronization problems (Producer-Consumer / Dining Philosophers) and identify the root cause of conflicts.	CO2	K4

		C) Demonstrate logical-to-physical address translation using a single-level page table. Explain how the page number and offset from a logical address are used to locate the corresponding frame in physical memory, and calculate the final physical address with an example.	CO3	K3
8	SATHRACK L SHANJU S SHARANESHWAR V	A) Show how a user command is converted into a system call in an operating system, using a file open or read operation as an example. Explain the steps from the user application request to the kernel execution, highlighting how the OS ensures controlled access to hardware resources. B) Compare mutexes and semaphores in terms of functionality, usage, and performance. Analyze situations where each is preferred, highlighting how they handle mutual exclusion, synchronization, and process coordination in concurrent programming. C) Apply the structure of a page table to demonstrate how a frame number is obtained during memory access. Explain the steps of translating a logical address into a physical address using the page number and offset, and show how the page table maps pages to frames.	CO1 CO2 CO3	K3 K4 K3
9	SHANMATHI R SHANMUGA PRIYA S SHARIKKA V	A) Apply the concept of CLI and GUI by explaining how a user performs the same task in both interfaces. B) Analyze the four necessary conditions for deadlock—mutual exclusion, hold and wait, no preemption, and circular wait—using a database transaction scenario. Explain how each condition can lead to deadlock and suggest strategies to prevent it. C) Apply segmentation to divide a program into logical parts such as code, data, and stack. Explain how each segment is mapped to physical memory and discuss the advantages of segmentation over contiguous memory allocation.	CO1 CO2 CO3	K3 K4 K3
10	SHYAMALA G SIVA SANKARI T SIVA RANJANI S	A) Apply the modular operating system structure concept to explain how device drivers can be added or updated without affecting the kernel. Discuss how modularity improves maintainability, flexibility, and system stability. B) Analyze and compare deadlock prevention, avoidance, detection, and recovery methods. Discuss their advantages, disadvantages, overhead, and practicality in real-world operating systems.	CO1 CO2	K3 K4

		C) Apply the concept of segmentation with paging to demonstrate how a large program is efficiently stored in memory. Explain how combining segmentation and paging reduces fragmentation, supports logical organization, and enables flexible memory allocation	CO3	K3
11	SINCHAN KUMARDEY P	A)Analyze the computer system components and explain how failure of one component affects overall system performance.	CO1	K4
	SIVAMANI K	B)Apply the process state diagram to explain how a music player application moves between states during execution.	CO2	K3
	SIVARAMAN V	C)Apply the concept of paging to explain how external fragmentation is eliminated in modern operating systems. Discuss how dividing memory into fixed-size pages and frames allows efficient allocation and simplifies memory management	CO3	K3
12	SNEHA M	A)Analyze the need for an operating system by comparing a computer with OS and without OS.	CO1	K4
	SOWMIYAR T	B)Demonstrate process creation and termination using a parent-child process example in a Linux system.	CO2	K3
	SRILEKHA S	C) Analyze the impact of limited main memory on system performance in a multiprogramming environment. Explain how insufficient memory affects CPU utilization, process execution, and overall system throughput.	CO3	K4
13	SUDHARSAN M	A)Compare and analyze Batch OS, Multiprogramming OS, and Time-Sharing OS based on efficiency and user interaction.	CO1	K4
	SURENDHAR M	B)Apply Inter-Process Communication (IPC) mechanisms to explain how a chat application exchanges messages.	CO2	K3
	SURYA J	C)Analyze the advantages and limitations of swapping in modern operating systems. Discuss how swapping affects CPU utilization, memory management, and overall system performance.	CO3	K4
14	SURYA S	A) Analyze why time-sharing systems are more suitable than batch systems in multi-user environments. Discuss the impact on responsiveness, CPU utilization, and user convenience.	CO1	K4
	TAMIL AMUDHAN U	B)Apply CPU scheduling criteria—waiting time, turnaround time, and response time—to a college admission system. Explain how these metrics affect the efficiency and fairness of processing multiple	CO2	K3
	TAMILVANAN G			

		student applications. C) Compare and analyze contiguous and non-contiguous memory allocation methods. Discuss their impact on memory utilization, fragmentation, and system performance.	CO3	K4
15	TARUN S THAMIZH PRIYAN V THIRUPATHI S	A) Analyze the advantages and limitations of monolithic and micro kernel architectures. B) Use First-Come, First-Served (FCFS) CPU scheduling to calculate the waiting time and turnaround time for a given set of processes. Analyze how the order of process arrival affects system performance and overall CPU utilization. C) Analyze the causes and effects of internal and external fragmentation.	CO1 CO2 CO3	K4 K3 K4
16	VASANTH S VELMURUGAN V VEMULA MOUNISH	A) Examine the OS services required for real-time applications like online gaming or video conferencing. B). Apply Round Robin (RR) CPU scheduling to explain how time-sharing works in a multi-user system. Illustrate how the CPU allocates fixed time slices to each process, how context switching occurs, and analyze the effect on responsiveness and fairness among users. C) Analyze why paging reduces external fragmentation but may increase internal fragmentation.	CO1 CO2 CO3	K4 K3 K4
17	VENKATESH P VIGNESH KUMAR C VINOTH KUMAR M	A) Analyze the role of system calls in providing security and protection in an operating system. Explain how system calls act as controlled gateways between user programs and hardware, enforce access restrictions, and prevent unauthorized operations, using real-world examples. B) Apply the concept of multi threading to explain how a web browser downloads multiple files simultaneously. C) Explain the relationship between I/O wait time and CPU utilization using multiprogramming in a college lab as an example.	CO1 CO2 CO3	K4 K3 K4

18	VISHAL V VISHVA J VISWANAATH A G	A) Compare Command-Line Interface (CLI) and Graphical User Interface (GUI) in terms of usability, efficiency, flexibility, and learning curve. Analyze their suitability for system administrators versus normal users, providing real-world examples of tasks best performed with each interface.	CO1	K4
		B) Demonstrate how mutex locks can be used to protect a shared variable during bank account transactions in a multi-threaded environment. Explain how mutex ensures data consistency when multiple users try to deposit or withdraw simultaneously, and analyze the impact on system reliability and performance.	CO2	K3
		C) Compare and analyze paging and segmentation with respect to protection and flexibility.	CO3	K4
19	YOGESH S YUGANANDHAN R YAASHIK H HARIHARAN S U	A) Analyze the importance of OS structuring methods in improving system reliability and maintainability.	CO1	K4
		B) Apply the concept of semaphores to control access to a printer shared by multiple users in a computer system. Explain how semaphore operations (wait and signal) can prevent conflicts and ensure mutually exclusive access, and analyze the impact on system efficiency and user experience.	CO2	K3
		C) Analyze how segmentation with paging combines the strengths of both memory management techniques.	CO3	K4
		D) Analyze the total memory access time in a system that uses paging by considering both the page table lookup and main memory access. Explain how techniques like TLB (Translation Lookaside Buffer) can optimize access time, and discuss the impact on system performance.	CO3	K4

20	VISHNUPRIYA P YAMIMA M YUVASHRI N	A)Analyze how design and implementation choices in an operating system affect performance, portability, and scalability.	CO1	K4
		B)Apply OS deadlock prevention techniques to a traffic intersection and explain how traffic rules prevent mutual exclusion, hold and wait, no preemption, and circular wait to ensure smooth traffic flow.	CO2	K3
		C). Analyze how different memory management techniques—such as paging, segmentation, and virtual memory—affect a computer system’s speed, security, and scalability. Use real-world analogies or scenarios to explain how these techniques improve system performance and protect data while supporting larger and more complex applications.	CO3	K4

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